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Atlas of Mercury

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(REV. 3-74)

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NASA SP-423

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Scientific and Technical
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1978

SEP 8 1980

Dedication

This Atlas is dedicated to the members of the Television Science Team. It was their efforts, started almost five years before the spacecraft reached Mercury, that made this Atlas possible. Their technical and scientific capabilities, coupled with their dedicated motivation, produced the high quality photographs included in this book. The photographs will be used by scientists the world over to study and understand the processes that have shaped the surface of Mercury.

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The Mariner 10 mission to Venus and Mercury scored many firsts. It was the first multiple-planet mission, borrowing energy from the gravity of Venus to make possible a flight to Mercury otherwise unachievable. This required navigation of a precision never before attempted—equivalent to shooting a rifle bullet through a 2-inch knothole more than 100 miles away. During its Venus swingby, Mariner 10 took the first close-up photographs of Venus, revealing the intricate spiral structure in its cloud layers that confirmed the classic circulation theory hypothesized by the astronomer Hadley more than 200 years ago and believed to be the basic driving mechanism behind weather on Earth. On the way from Venus to Mercury, Mariner 10 also made the first practical use of solar sailing, a novel technique that I predict will be used increasingly in the future to replace more expensive space propulsion systems. And as it flew by Mercury, Mariner 10 entered an orbit that, for the first time, provided two subsequent flyby revisits.

The hard-working Mariner 10 team also scored a number of management firsts. It was the first space project team to ever receive a NASA performance award *prior* to launch—a tribute to their determination and skill in pioneering daring techniques to cut the cost of space missions at the same time they were actually upgrading the quality of the science return.

Of all the firsts, undoubtedly the outstanding achievement of the Mariner 10 mission was the spectacular unveiling of the planet Mercury. Mercury's closeness to the Sun makes it an almost impossible object for astronomical study, and the total knowledge of Mercury prior to Mariner 10 was miniscule. Even its rate of rotation was not determined until 1965. Mercury's surface was almost totally unknown, with considerable conjecture that total surface melting could have left Mercury as smooth as a billiard ball.

Then came the tiny Mariner 10 spacecraft, a bright sunlit speck of a solar sailing craft speeding in from the

blackness of space. Aboard was an imaging system born of the ingenuity and close cooperation between space systems engineers and the scientists of the imaging experiment team. Using a narrow-angle television camera, it could take only postage-stamp-size pictures of the surface. But it could flash them back to Earth with such rapidity that it was possible to map the entire lighted portion of the planet with excellent resolution.

This Atlas is a tribute to the accomplishments of that highly productive team effort. As you turn its pages you will see the face of Mercury as it was unveiled to mankind for the very first time. If it is not a beautiful face, it is nevertheless a most fascinating one, marked with a character all its own, including "wrinkles" over 2000 km long. Even its noticeable similarities to the Moon are fascinating—why should both the Moon and Mercury have the smooth mare areas located predominantly on one face with rough highlands on the other? Why should Mercury, so far from the asteroid belt, have a surface just as pocked by bombardment as the Moon? Clearly, adding the portrait of Mercury to our gallery of terrestrial planets will contribute greatly to our knowledge of the violent accretion process that formed the planets.

Even with the two revisits of Mariner 10 to Mercury, we have seen only one side of the planet—a limitation imposed by Mercury's harmonic rotational lock to the Sun. I wait with eager anticipation for the day when we return to see the other face of Mercury.

Robert S. Kraemer
Director, Lunar and Planetary Programs
National Aeronautics and Space Administration
October 1976

Foreword

counter phases. The schematic view of the television camera is shown in Figure 7, and the camera characteristics are given in Table 3.

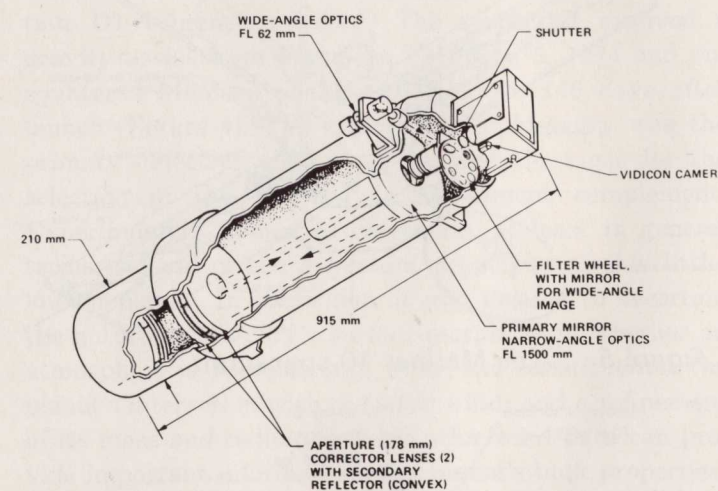


Figure 7 Schematic view of Mariner 10 television camera.

The imaging sequence was initiated 7 days before the encounter with Mercury when about half of the illuminated disk was visible and the resolution was better than that achievable with Earth-based telescopes. Photography of the planet continued until some 30 min before closest approach, providing a smoothly varying sequence of pictures of increasing resolution and decreasing areal coverage. Pictures with resolutions on the order of 2 to 4 km were obtained for both quadratures on the first encounter (Figures 18 and 19). Variation in resolution, ranging between several hundred kilometers to approximately 100 m, assisted in the extrapolation of large-scale features observed at high resolution over broad areas photographed at lower resolution. The highest resolution photographs were obtained approximately 30 min prior to and following closest approach on the first and third encounters. Pictures taken in a number of spectral bands enabled the determination of regional color differences.

The second Mercury encounter (Figure 6) provided a unique opportunity to observe regions of Mercury with more favorable viewing geometry than was possible during the first encounter. In order to permit a third encounter, it was necessary to target the bright-side encounter for a south polar pass. This trajectory allowed unfore-shortened views of the south polar region, the exploration of areas not previously accessible for study, a geologic and cartographic tie in the southern hemisphere between the two sides of Mercury photographed on the first encounter,

and the acquisition of stereoscopic coverage of the southern hemisphere. Because of the small field of view resulting from the long focal length optics, it was necessary to increase the periaapsis altitude to about 48,000 km to ensure sufficient overlapping coverage to make a reliable geologic and cartographic tie. The resolution of the photographs taken during closest approach ranged from 1 to 3 km (Figure 20).

Table 3 Television Camera Characteristics

Focal length	1500 mm (62 mm) ^a
Focal number	f/8.4
Shutter speed range	33.3 ms to 11.7 s
Angular field of view	0.38° × 0.47° (9° × 11°) ^a
Vidicon target image area	9.6 × 12.35 mm
Scan lines per frame	700
Image elements per line	832
Bits per image element	8
Frame time	42 s
Spectral filters	Blue, ultraviolet, ultraviolet polarizing, orange, minus ultraviolet, and clear

^aWide-angle optics.

The third Mercury encounter was targeted to optimize the acquisition of magnetic and solar wind data. Therefore, the viewing geometry on the third encounter was very similar to that on the first encounter. However, the third encounter presented the opportunity to target high-resolution pictures to areas of geologic interest seen previously at lower resolution. Because of ground communication problems, these pictures were acquired as quarter frames.

Infrared Radiometry. The primary goal of the infrared radiometry experiment was to measure infrared thermal radiation emanating from the surface of Mercury between late afternoon and early morning. These temperature measurements taken on the first encounter provided much more accurate values for the average thermal properties of the planet than can be obtained from ground-based studies. An important secondary objective was to search for possible correlations between thermal anomalies and topographic features.

Ultraviolet Spectroscopy. The occultation spectrometer provided a sensitive detection of any atmosphere present, and of its composition, with a detection threshold improved by a factor of about 10^7 over current ground-based studies. The airglow spectrometer provided quantitative information on the abundance of H, He, He⁺, C,

O, Ne, and A in the the intensity and s emission lines. Dat encounters.

Celestial Mech tial mechanics expi tments of the mass Mercury. The plane orbital eccentricity made this experimer

The occultation c first encounter affo mosphere and to me changes in the S-ba of an atmosphere w more sensitive but l

MAP PROJECTIONS

Many different projections are used in making maps; the choice depends on the purpose of the map and the type of distortions which can be tolerated.²² Some form of distortion is always present when a sphere or spheroid is mapped into a plane, and the selection of the best projection for a particular cartographic product must reflect a compromise of the allowed distortions and the use of the map. Most map projections are designed to give a proper representation of distance (equidistance), shape (conformal), or area (equivalence); however, a projection cannot possess more than one of these properties.

There are three common projection surfaces—the cylinder, the cone, and the plane. Normally the cylinder is tangent to the sphere at the equator; sometimes, however, the transverse or oblique positions are used. With the transverse position the line of tangency is at a selected meridian; with the oblique position the line of tangency is at an angle to the equator and all meridians. When a conical surface is used, it is generally either tangent to the sphere along a particular latitude or it cuts the sphere along two lines of latitude. When a plane is used as a projection surface, it is usually tangent to the sphere at a single point such as the north or south pole.

Over the centuries a great many projections have been devised and employed in making maps of the Earth and its many regions. Land-water boundaries, political areas, roads, or cities are frequently of primary interest. For other planets, these considerations are irrelevant and the center of interest is mainly in the topographic forms and positions. Because it is important to represent accurately the shapes of the topographic features, the map projection should be conformal. Computers are frequently used to project a picture or mosaic into a map. If the projection is conformal, the craters will be round, thus providing a check on the computer program. All of the maps in this Atlas use conformal projections.

The most popular cylindrical projection is the Mercator, which is conformal; the cylinder is usually oriented tangent to the reference sphere at the equator. The transverse Mercator is becoming increasingly popular for Earth cartography and, together with oblique Mercators, will likely find application on other planets. The Lambert normal conical projection is conformal and is useful with one or two standard parallels in the midlatitudes. The stereographic plane projection is conformal and is commonly used in the polar regions with the point of tangency at the pole. Occasionally this projection has the point of

tangency at the equator. It has recently been exploited in special maps of large basins found on the Moon, Mars, and Mercury.

NOMENCLATURE

Since the time of Schiaparelli, a number of astronomers have drawn maps of the surface markings on Mercury; however, only Lowell (1896)²³ and Antoniadi (1934)¹ gave names to the features on their sketches. Lowell's map is shown as Figure 9 and Antoniadi's is presented as Figure 10. To the extent that nomenclature was used prior to the flight of Mariner 10, Antoniadi's was generally accepted.

At the 1973 meeting of the International Astronomical Union, a Working Group for Planetary System Nomenclature was established. Recommendations made by the Task Group for Mercury Nomenclature²⁴ must be ap-

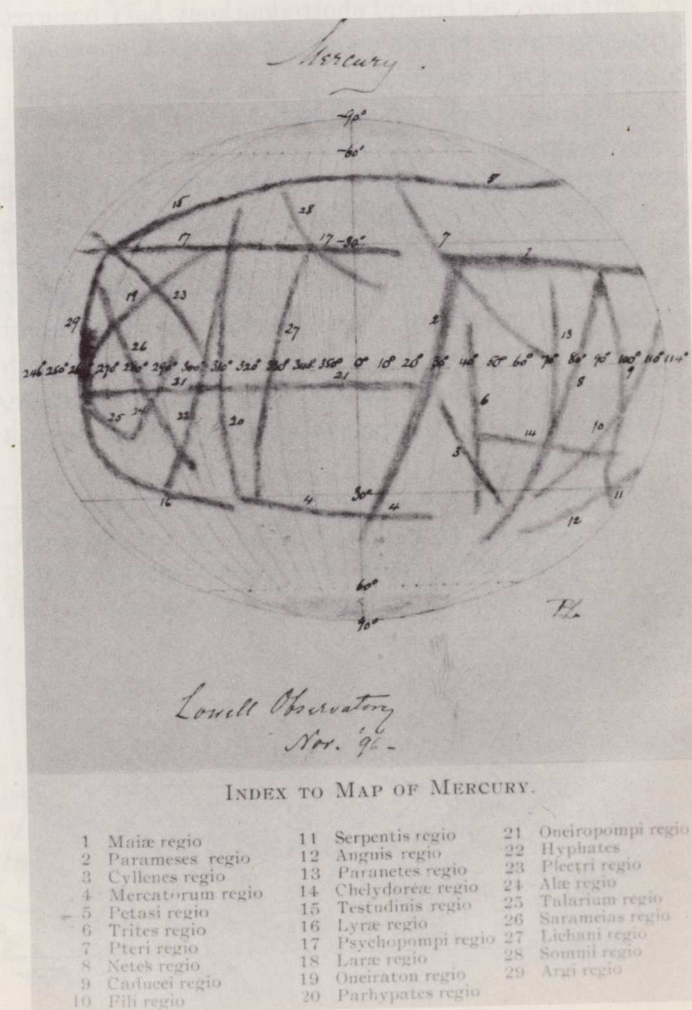


Figure 9 Lowell's map of Mercury.²³

proved first by the Working Group and then by the Executive Committee of the International Astronomical Union.

For the convenience of telescopic observers, the names of albedo features shown in Figure 11 have been adopted by the Task Group from names originally given by Antoniadi to markings on the surface of Mercury. The rela-

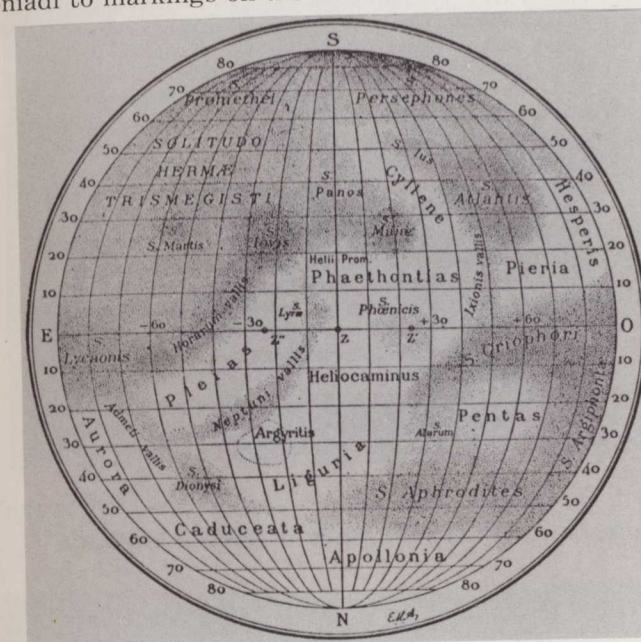


Figure 10 Antoniadi's map of Mercury.¹

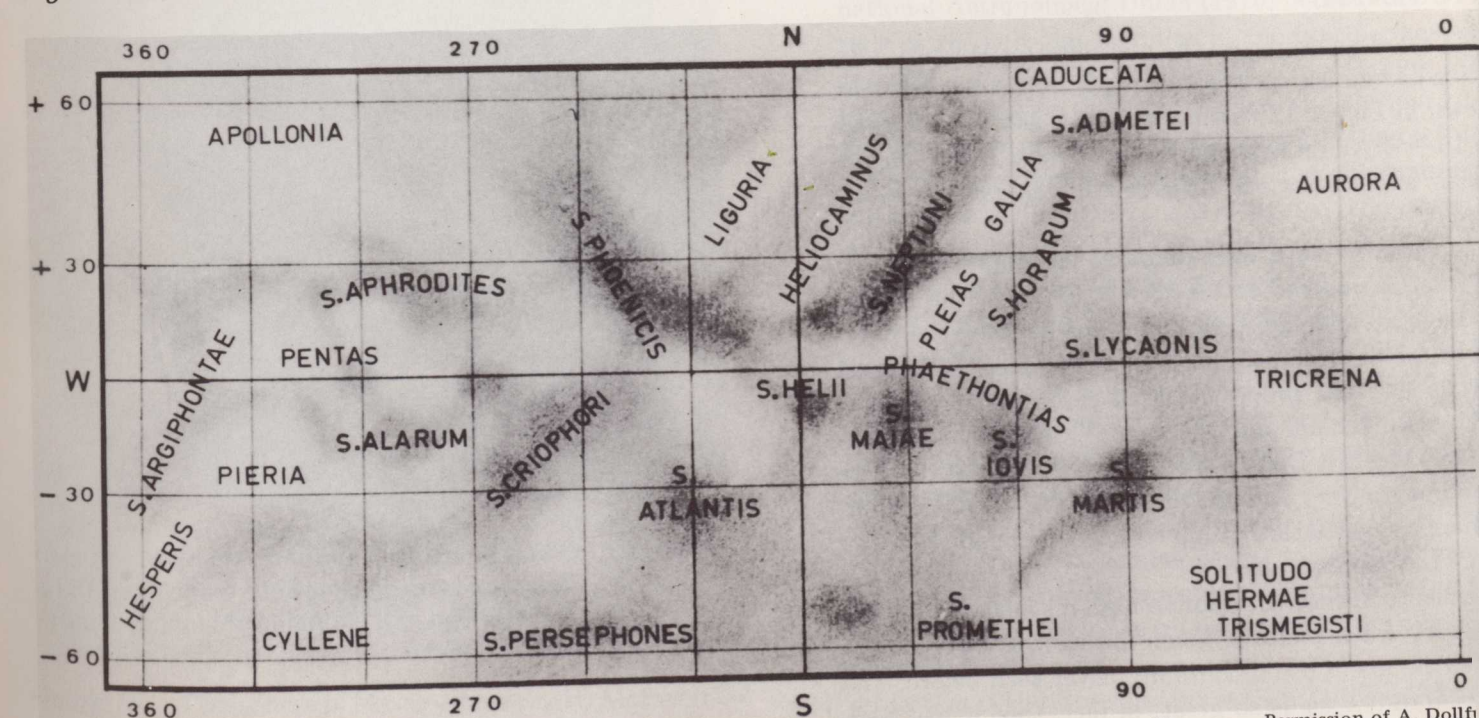


Figure 11 Names of albedo features adopted by the International Astronomical Union.²⁵

Permission of A. Dollfus
Observatoire de Paris—

tionship of the markings to the topography is very different from that on the Moon, where the dark markings correspond to the maria—the large flooded basins. On Mercury, the albedo variations seem to be due to the brightness of the extensive ray systems, because the albedos of the large flooded basins do not differ greatly from those of the surrounding cratered terrain.

The assignment of new names to topographic features is a continuing activity, with additions made as users require them. Maps are commonly used to locate and identify named features. Although cartographers are the primary source for requests for new names, photogeologists working with pictures and maps also require names for important features. Thus, the Task Group for Mercury Nomenclature must maintain close contact with scientists actively studying Mercury in order to supply names as required.

The large craters on Mercury are named after authors, artists, and musicians. Typical names are Homer, Rembrandt, and Bach. Two exceptions to this general rule are Kuiper and Hun Kal. Kuiper is named after Dr. Gerard Kuiper of the University of Arizona, a member of the original Mariner Venus/Mercury Imaging Team who died in Mexico City in December 1973 before Mariner 10 reached Venus. The crater Kuiper is located at 11°S and 31°W, 60 km in diameter, and has an extensive bright ray

Description

For cartographic purposes, Mercury has been divided into 15 geographical regions (Figure 17). The Mariner 10 television cameras were able to take pictures of the planet's surface corresponding approximately to only 9 of these 15 regions because the same hemisphere was illuminated during all three encounters as a consequence of the synchronous nature of Mercury's rotation and the orbit of the spacecraft around the Sun. This Atlas is divided into 9 sections, each representing one of the cartographic regions. The name and H (prefix for Mercury) number of each region are shown in Figure 17. The regions presented are H-1, H-2, H-3, H-6, H-7, H-8, H-11, H-12, and H-15.

All sections of the Atlas are arranged in the following manner: A 1:5,000,000 shaded relief map and a computer-generated photomosaic are presented first on facing pages for general reference. Subsequent material includes enlargements of portions of the photomosaics, individual high-resolution pictures, mosaics of small areas, and stereo pairs located within the boundaries of the cartographic region. The photomosaics are designated by the letter A for the 1:5,000,000 format and the letters B, C, ... for the enlarged versions (e.g., 1-A, 1-B, ...). Individual pictures, small mosaics, and stereo pairs are designated by a numerical identification (e.g., 1-1, 1-2, ...), where the first digit denotes the cartographic region and the second digit is its identification within the region.

Footprint locations of individual pictures and stereo pairs have been plotted on the shaded relief maps. In cartographic regions H-1 and H-2, where high-resolution, third-encounter photographs were obtained on the planet's limb, footprints are provided both on the limb mosaic and on the shaded relief map. Footprint maps are identified by a 3 unit symbol (e.g., 1-F1, 1-F2, ...), where the first digit denotes the cartographic region and the last two symbols are its identification within the region.

The shaded relief maps are adapted from the 1:5,000,000 series rendered by airbrush artists in the Cartographic Section, Branch of Astrogeological Studies, U.S. Geological Survey (USGS), Flagstaff, Arizona. These maps show the topography without reference to albedo or sun direction, factors which combine to produce a surface appearance often quite different from that in the pictures. Names of surface features and the latitudes and longitudes on the maps make the first page the major reference for each section.

All photo products were produced using computer techniques and software developed in the Image Processing

Laboratory (IPL) of the Jet Propulsion Laboratory (JPL), California Institute of Technology. The pictures have been high-pass filtered and contrast enhanced to accentuate surface detail. Pictures used in photomosaics or in stereo pairs have been geometrically transformed to an appropriate projection. A picture element size of 0.4 km was used in the mosaics and a video film converter was used to record the entire photomosaic on 8 by 10 in. film to preserve the resolution inherent in the photographs.

During the first flyby encounter, full coverage of both hemispheres was obtained with a resolution of about 2 km. Individual pictures were processed by IPL and mosaicked by USGS to produce Figure 18, a view of Mercury as seen from the approaching spacecraft, and Figure 19, a view of Mercury as seen from the departing spacecraft.

The second encounter, 6 months later, was on the illuminated side of the planet. Twenty sequences, composed of 18 pictures each, yielded coverage of an area extending from the equator to the south pole and encompassing the south portions of both hemispheres photographed earlier. The pictures were obtained at rapidly changing ranges and viewing angles, thus preventing the combination of the raw photographs into a global mosaic. Therefore, each frame was processed by IPL as an orthographic projection with an origin at -55° latitude and 100° longitude. Figure 20 is a mosaic of these pictures and provides a global view of Mercury as seen by the spacecraft as it passed above the point of projection. The only illuminated area not photographed during the three encounters was north of the equator between the limb coverage of the first encounter. This "gore" is especially evident in the H-2 region.

Pictures taken during the third encounter, 1 year after the first visit to the planet, are identified by their strip-like configuration (see, for example, picture 1-13). Very-high-resolution pictures were obtained in specific areas of interest identified during the first pass.

The Caloris basin, bisected by the morning terminator, is the most prominent feature discovered on Mercury. Because of its importance, a special mosaic, Figure 21, was constructed at JPL from the highest resolution pictures of the area obtained during all three encounters. Each photograph was scaled to a similar proportion and additional enhancement matched tone contrast from one picture to another. To eliminate foreshortening and suppress picture edge effects inherent in Figure 21, a stereographic projection was made of the Caloris basin (photomosaic 3-F).

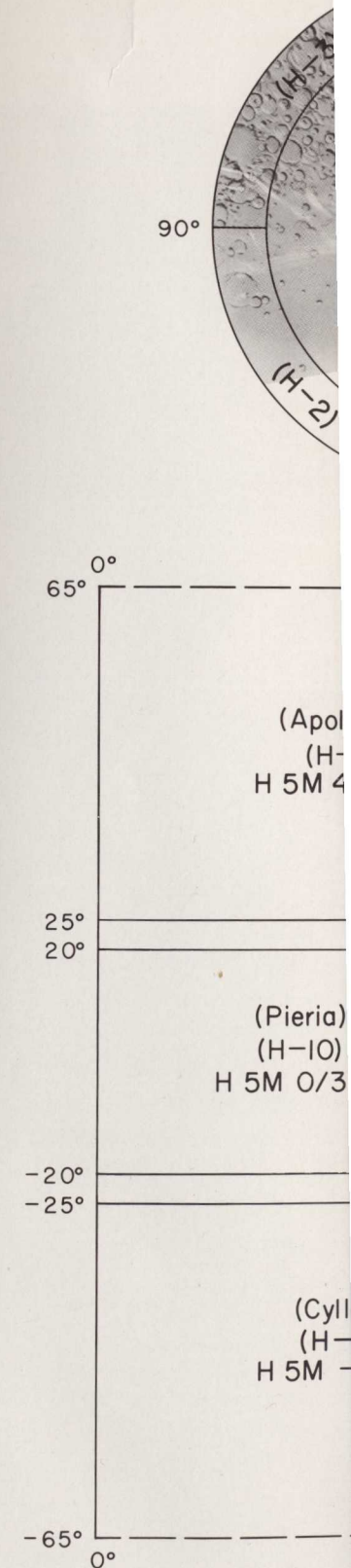
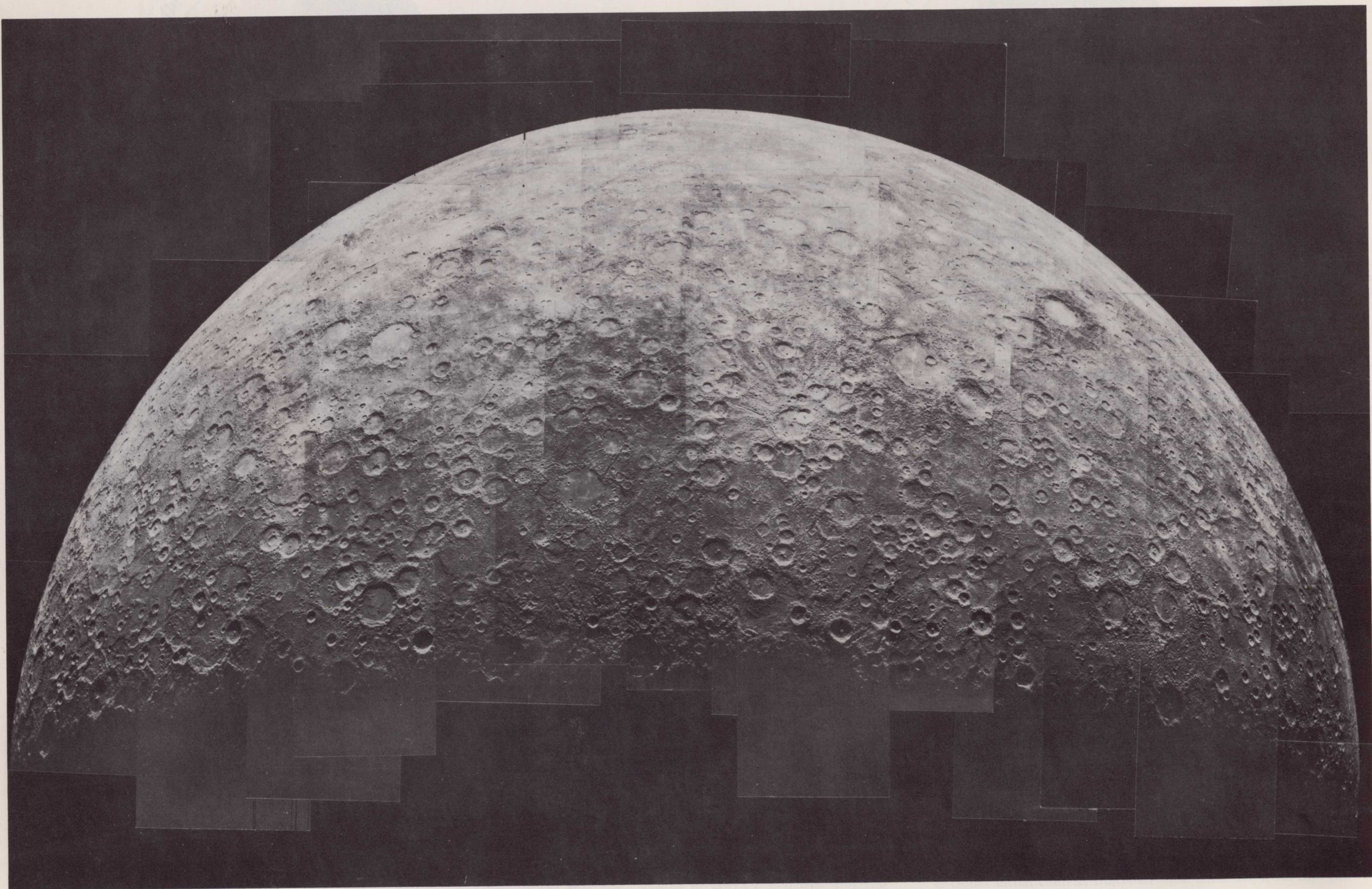
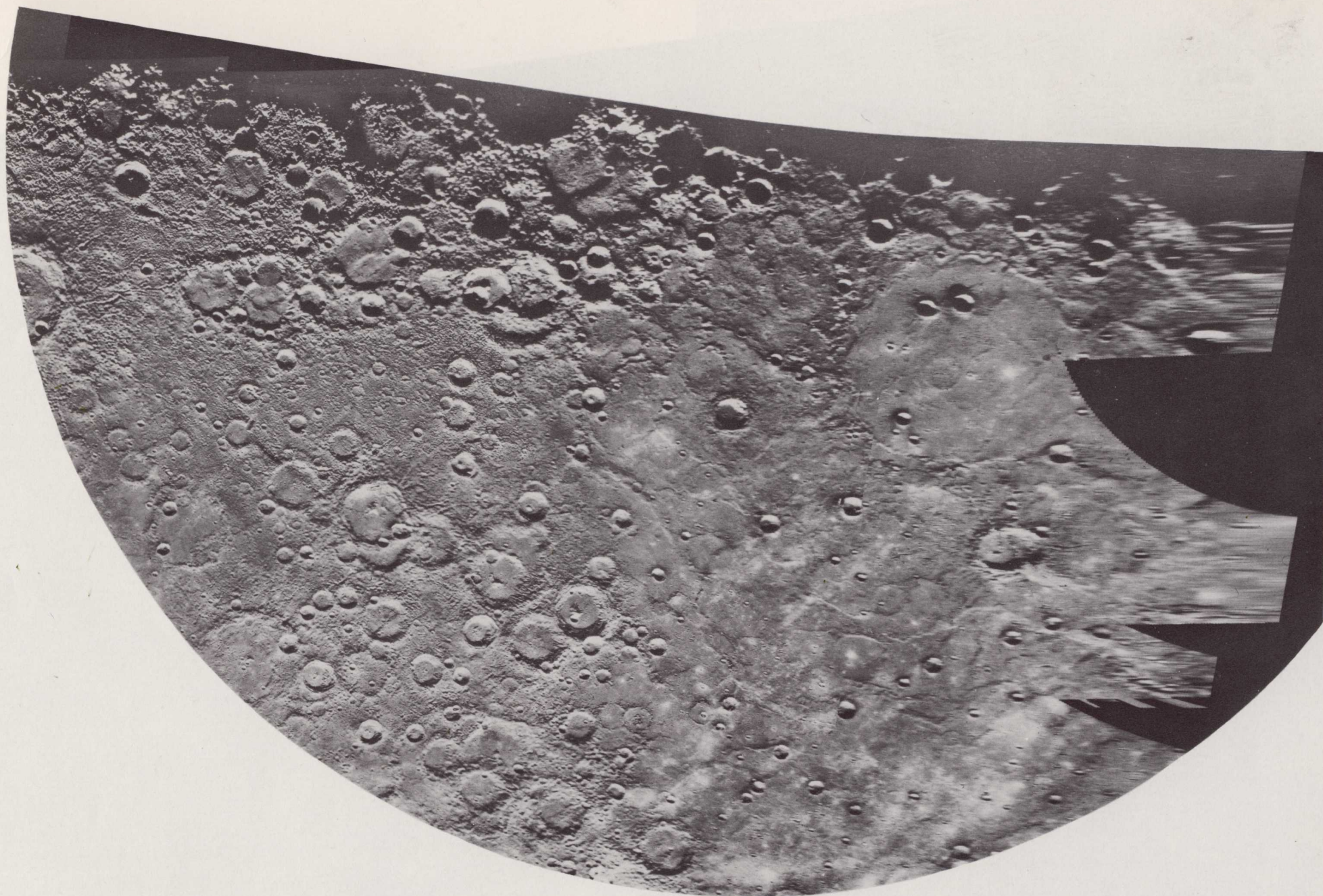


Figure 18 Photomosaic of Mercury as photographed by the approaching spacecraft





1-A COMPUTER PHOTOMOSAIC OF THE BOREALIS AREA OF MERCURY

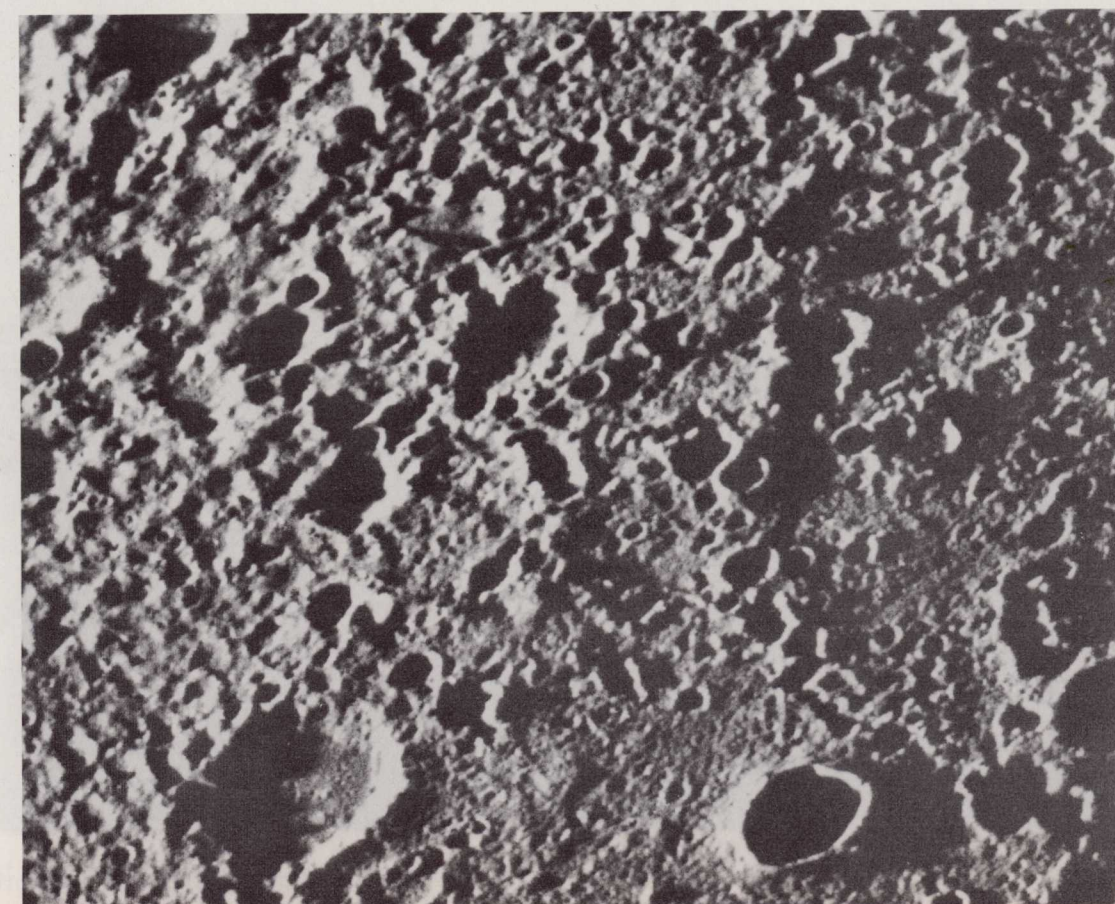
H-1



6-F1 Footprints of pictures 6-1 through 6-11 and 6-14 on the shaded relief map.



6-1

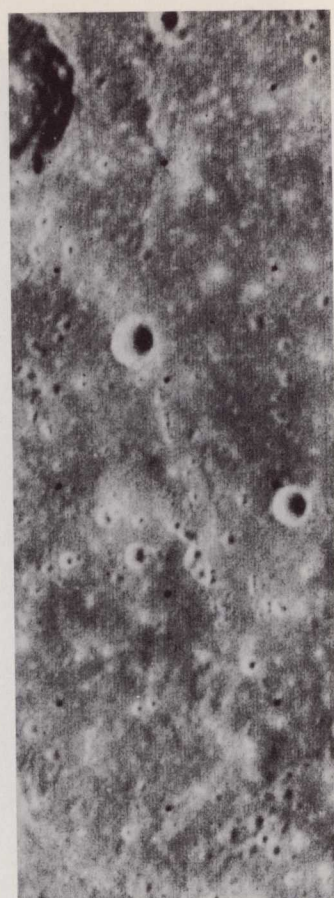


6-3

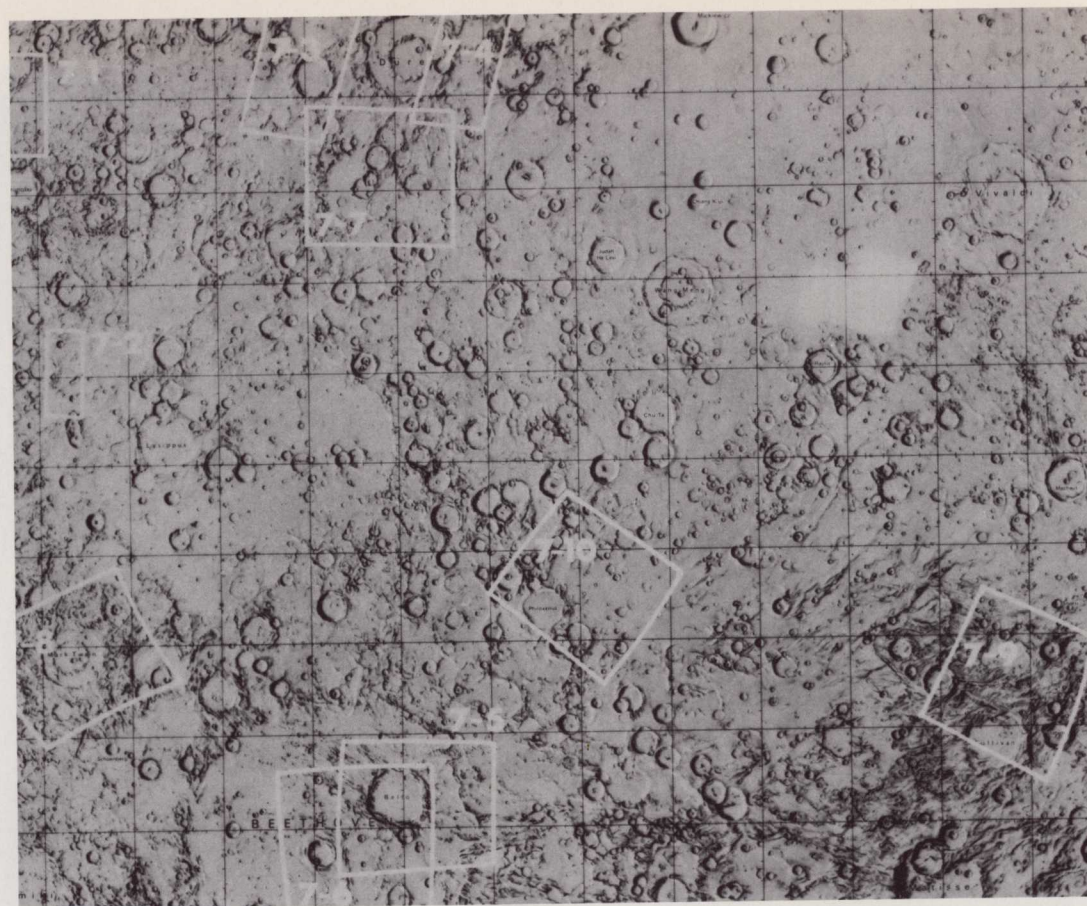
6-2



7-1

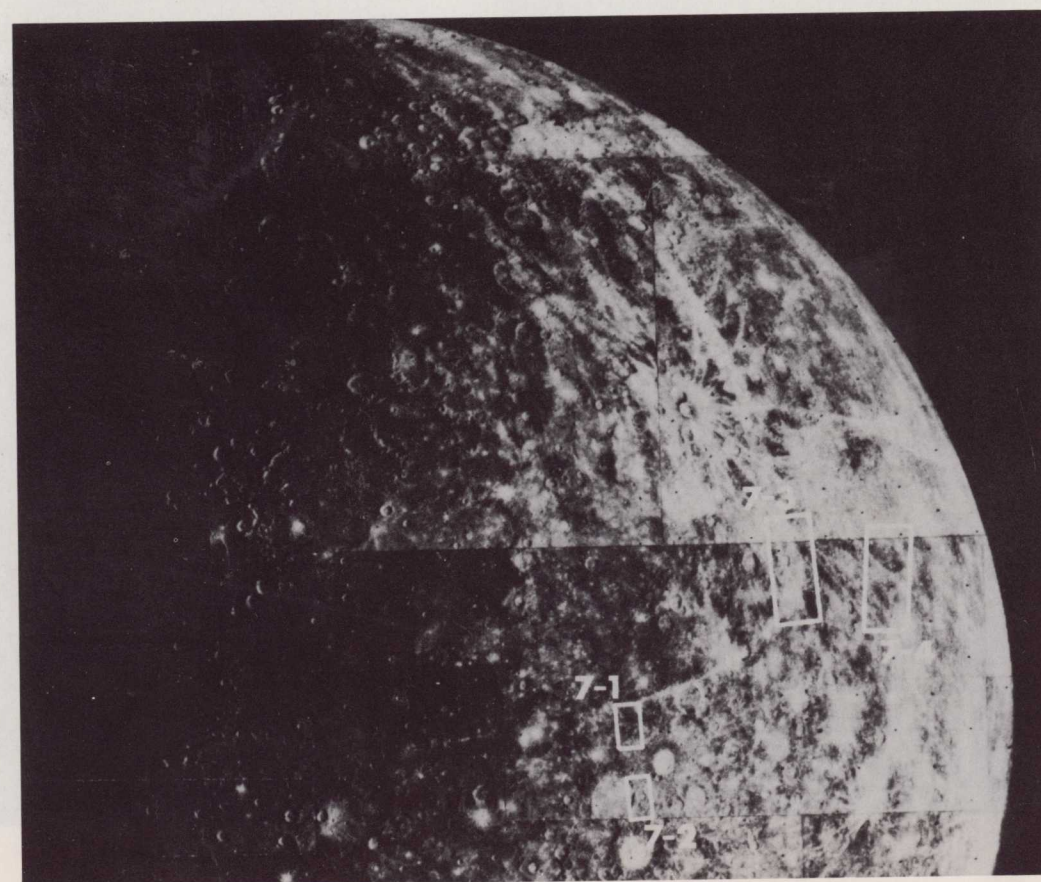


7-2



7-F1 Footprints of pictures 7-1 through 7-10 on the shaded relief map

7-3



7-4

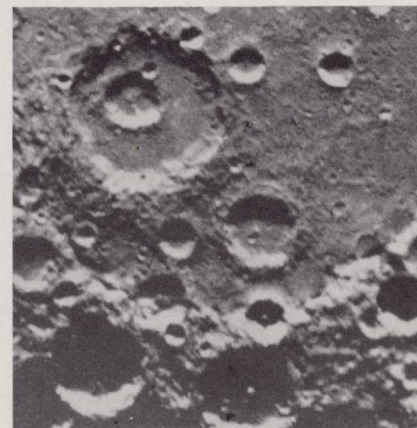
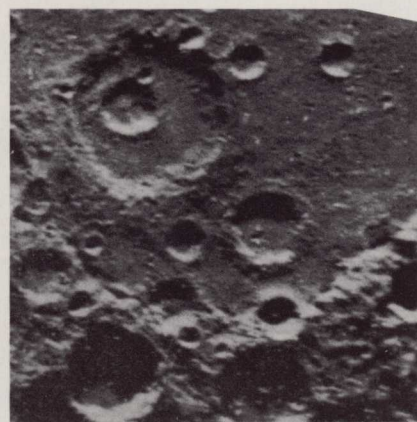


7-F2 Footprints of pictures 7-1 through 7-4 as they appear on the limb

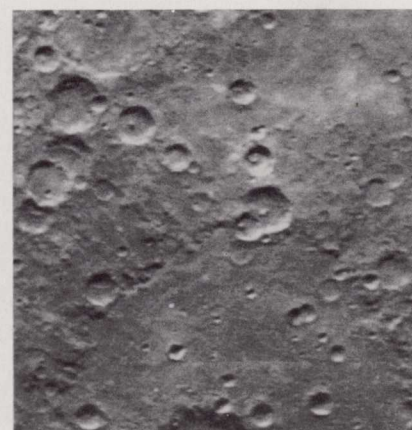


12-F1 Footprints of stereo pairs 12-1 through 12-12 on the shaded relief map

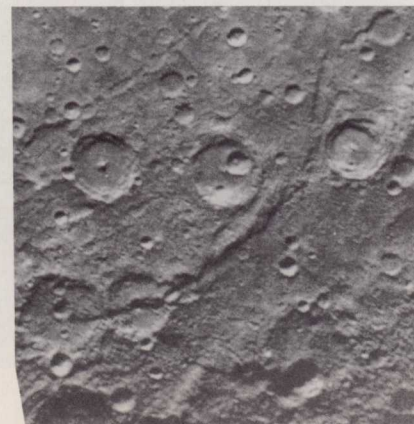
12-1



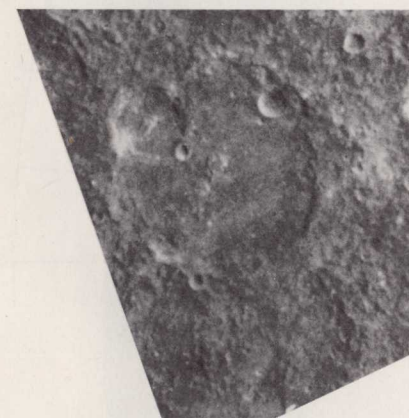
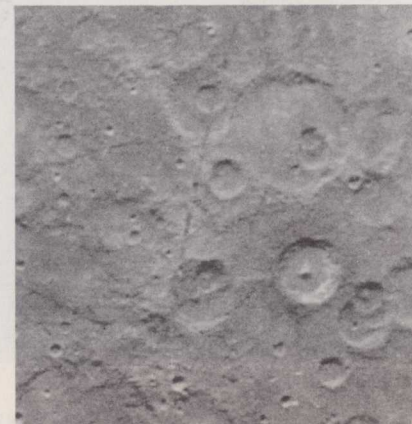
12-3



12-2



12-4



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THE **M** ATLAS OF
MERCURY

CHARLES A. CROSS & PATRICK MOORE

Mercury within the Solar System

Behaviour and characteristics before and after Mariner

Since the observational work of Antoniadi and his contemporaries, new techniques, especially the use of radar, have added to the basic knowledge of Mercury, which Mariner confirmed. Thus the value used by Antoniadi for the mean distance of Mercury from the Sun at 35,950,000 miles (57,850,000 km) has been revised to 35,980,000 miles (57,910,000 km). However, his eccentricity value of 0.206 for Mercury's orbit still stands; there is a considerable range between the perihelion and aphelion distances (the closest and farthest points from the Sun). The orbit is more eccentric than that of any other planet apart from Pluto, and thus has a marked effect upon surface temperatures.

A small telescope will suffice to show the phases; it is worth noting that Mercury, unlike Venus, is actually brightest when gibbous—that is, at the phase between half and full. The fact that surface markings are so difficult to observe from Earth is due partly to the fact that Mercury is never seen against a dark background, and partly to the small size of the planet. The equatorial diameter is 3,030 miles (4,880 km). This is intermediate between the diameters of the Moon (2,159 miles/3,475 km) and Mars (4,220 miles/6,790 km). The escape velocity is a mere 2.7 miles (4.3 km) per second, so that Mercury would not be expected to retain much in the way of atmosphere.

Sizes compared

Mercury (A) is the smallest of the group of four inner planets; the other members are Venus (B), Earth (C) and Mars (D). Its equatorial diameter is 3,030 miles (4,880 km). Jupiter (E), the largest of the giant outer planets, is virtually 30 times greater in diameter; the Sun (F) more than 285 times greater. Before Mariner, measurements of Mercury's diameter, based on telescopic observation, were not exact. Original estimates varied from between 3,100 miles (5,000 km) and 2,900 miles (4,700 km). Even with the world's largest telescopes no object smaller than 500 miles (800 km) can be resolved.

An iron-rich core bigger than the Moon

On the other hand Mercury is very dense. Its specific gravity is about the same as that of the Earth: 5.5 (that is to say, Mercury "weighs" 5.5 times as much as an equal volume of water would do). This suggests a heavy, iron-rich core, much larger relatively than that of the Earth. The diameter of the core has been estimated at 2,200 miles (3,500 km), making it larger than the whole globe of the Moon. One interesting result of so high a density is that the surface gravity on Mercury is virtually the same as that on Mars; that is, 0.38, where Earth is the unit of 1. The mass of Mercury is 0.055 and the volume 0.056 of that of the Earth.

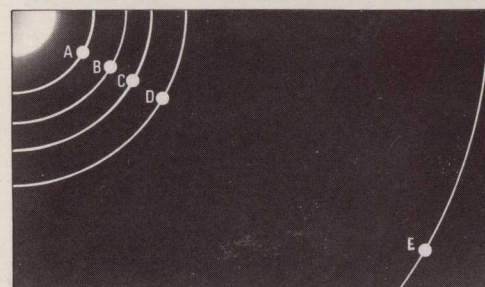
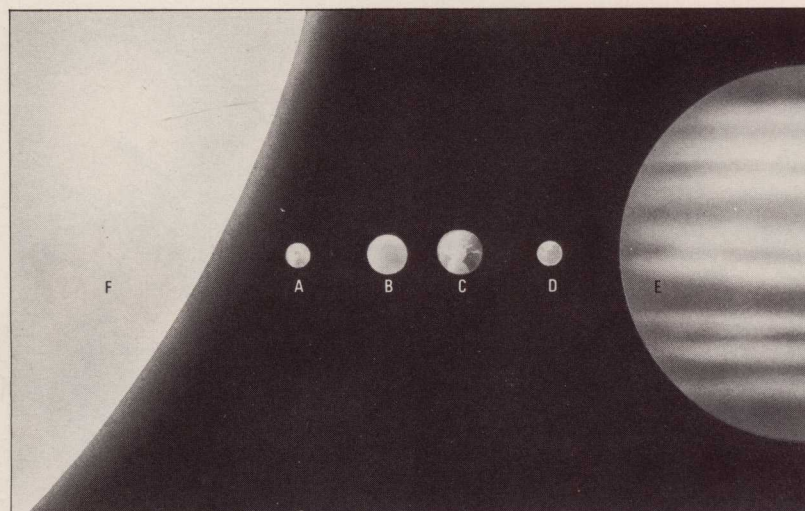
Mercury has an orbital period of 87.969 Earth-days. The belief of the early observers that the axial rotation period was synchronous was not challenged until 1962. At Michigan, in the United States, W.E. Howard and his colleagues measured the long-wavelength radiations coming from Mercury, and found that the dark side was very much warmer than it should be if it received no sunlight at all.

Determining axial rotation and surface depressions

The next developments came from radar work. Basically, this involves sending out a radio pulse, "bouncing" it off a solid body and recording the "echo". When pulses are reflected from a rotating body, the "echo" is affected, and the rate of spin can be found. This was done for Mercury, initially by R. Dyce and G. Pettengill, using the huge radio telescope at Arecibo, in Puerto Rico. They found that the axial rotation period was not synchronous; the actual period is 58.65 Earth-days, or two-thirds of the orbital period. There is no area of permanent daylight or permanent night and no twilight zone.

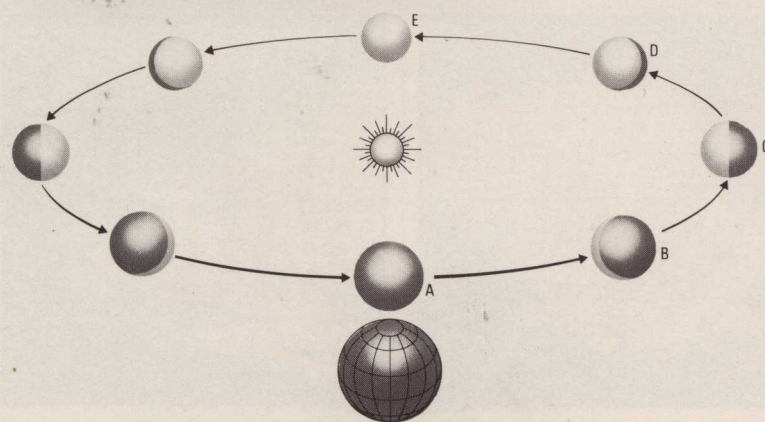
There was also a possibility of using radar to detect elevations and depressions on Mercury. Intensive studies were carried out at the Deep Space Station at Goldstone, California, in 1972. The results were interpreted by S. Zohar and R.M. Goldstein, who described the central regions of the planet as being "dotted with hills and valleys, plus apparent craters 30 miles (50 km) across here and there". It was then thought that the hills were of a gently undulating nature. It was not until the flight of Mariner 10 that the lunar-type surface was revealed.

From time to time searches have been made for a satellite of Mercury, but it now seems unlikely that any exist. On the other hand there have been suggestions that Mercury used to be a satellite of Venus; certainly it is true that in size the ratio of Earth to Moon is not so very different from that of Venus to Mercury. However, the theory is still highly speculative.



Distances compared

The distance of Mercury from the Sun ranges between 28.5 million miles (45.9 million km) at perihelion and 43.3 million miles (69.7 million km) at aphelion. The mean distances of the four inner planets (A–D) are shown to scale and compared to the distance of the first of the giant planets, Jupiter (E). (The Sun is not to scale.)

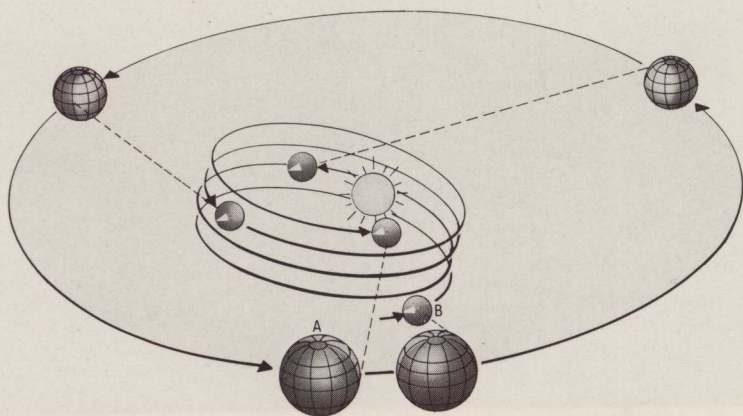
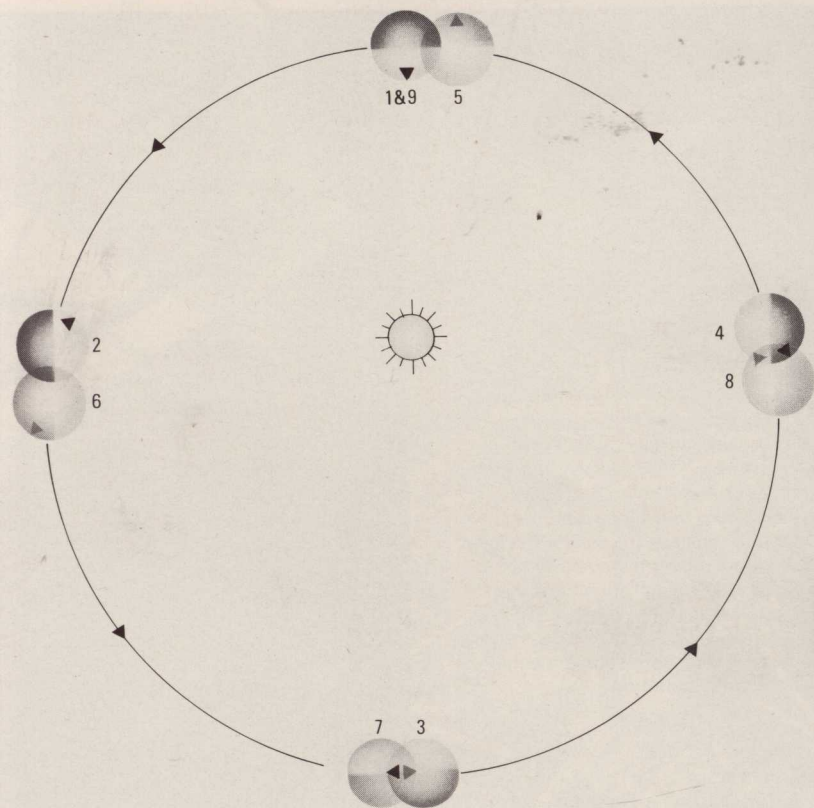


The phases of Mercury

When Mercury is between the Sun and the Earth it is new (at inferior conjunction) (A) and can only be seen in transit. As the planet moves along, a little of the daylight side begins to tilt in the Earth's direction; Mercury reaches a crescent (B), then a half (dichotomy) (C) and then the stage between half and full (gibbous) (D). At superior conjunction Mercury is full (E), but as it is then virtually behind the Sun it is barely observable. The mean synodic period is 115.9 Earth-days; that is, it takes Mercury just under 116 days to return to inferior conjunction.

Synchronous rotation disproved

Radar measurements, fully confirmed by the Mariner 10 results, show that the axial rotation period of Mercury is two-thirds of the orbital period. Mercury is shown with a fixed arrow indicating the direction of one of the two "hot poles"—that is to say, the points on Mercury's equator that directly face the Sun at perihelion. The numbers give the appropriate positions of the planet in its orbit during two revolutions around the Sun. Placed end to end, it is clear that Mercury has made three revolutions in two orbits. Each position is separated by an interval of 22 Earth-days. Both spin and orbital rotations are anticlockwise.



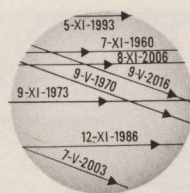
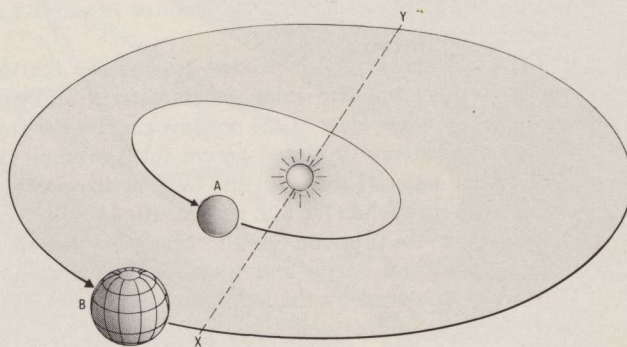
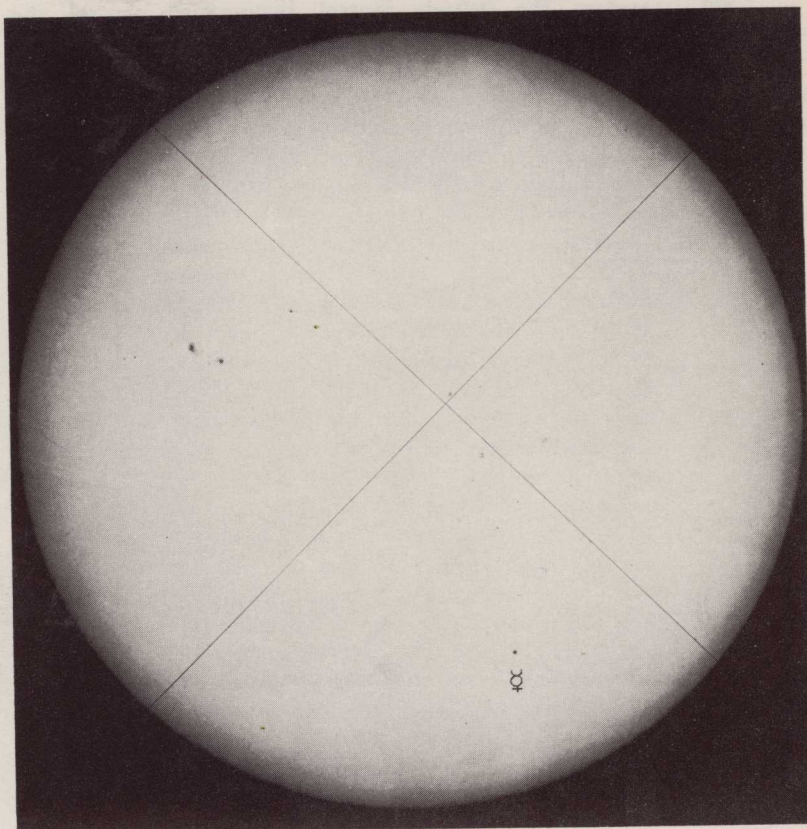
Mercury's elusiveness explained

The impression from Earth that Mercury keeps only one hemisphere turned towards us is explained by the planet's synodic period—that is, the time which elapses between successive appearances at the same phase. Because it takes Mercury two orbits to complete three revolutions, the extra revolution gives rise to a Mercurian day which is two years, or 176 Earth-days, long. This interval is approximately equal to $1\frac{1}{2}$ synodic periods, so that after every three synodic periods the same face of Mercury will be seen at the same phase. Three synodic periods of Mercury add up to approximately one Earth-year,

so that the most favourable times for observing the planet recur every three synodic periods—when the same hemisphere is turned towards us. The agreement is not exact, but it was near enough to mislead all astronomers into believing in a synchronous rotation, which was not disproved until the development of radar techniques during the 1960s. The diagram shows Earth (A) and Mercury (B) over three synodic periods. Mercury's revolution is indicated by the marker. The dotted line represents the view of an observer on Earth standing at the terminator (the boundary between the day and night hemispheres of the planet).

Transit characteristics

During a transit, Mercury, unlike Venus, cannot be seen with the naked eye. Telescopically it appears completely black and is much darker than any sunspots that happen to be visible. The comparison is clear in the photograph of the transit of Mercury that occurred on 7 November, 1914. During the eighteenth century it was suggested that transits of Mercury might, like those of Venus, be used to measure the length of the astronomical unit. Attempts were made to use the 1753 transit, but the results proved to be disappointing. Transits of Mercury are now regarded as being scientifically unimportant.



A fleeting alignment

Because Mercury's orbit (above) is inclined at an angle of 7° to the ecliptic, only during relatively short periods is there any chance that the Sun, Mercury (A) and the Earth (B) will become aligned when Mercury is at its inferior conjunction. Transits can occur only during the months of May (x) and November (y). Although transits of Mercury are not common, they are more frequent than those of Venus. The tracks of Mercury across the Sun during the transits from 1960 to 2016 are shown (left). Clearly the transit of 1986 will last longer than the succeeding one of 1993.

Mariner 10—the anatomy of a spacecraft

The use of gravity to alter orbits

Mariner 10's journey to the inner planets represented a pioneering achievement in space technology. Although building on techniques established in previous Mariner missions, Mariner 10 was the first vehicle to make use of the gravity of another planet—Venus—to gain its correct trajectory. It was also the first mission to transmit its television pictures as they were taken. Both the cameras and the more refined sensors incorporated on the vehicle provided a wealth of scientific information.

The design, development and construction of the whole vehicle was completed in 2½ years. The assembly is shown nearing completion, below, in good time for the launch date in November 1973. The facilities required to maintain the equipment discussed in the diagram were the propulsion system, the attitude control, temperature control, the power supply and the communications system.

Propulsion and attitude control

Because Mariner used the gravitational fields of both Venus and Mercury at each encounter to provide the main changes in direction and velocity, the propulsion unit was required to make only fine adjustments. The system provided a total velocity change of 390 ft (119 m) per second, and could be started and stopped as often as needed.

As the vehicle continued on its trajectory, it had to maintain a fixed attitude in space, with its Z axis pointing at the Sun, and its X and Y axes diagonal to the ecliptic plane. With this stable attitude, the sunshade protected the delicate instruments from the searing heat of the direct solar radiation, and the scientific instruments and the antennae were correctly pointed to make their measurements and transmit these back to Earth.

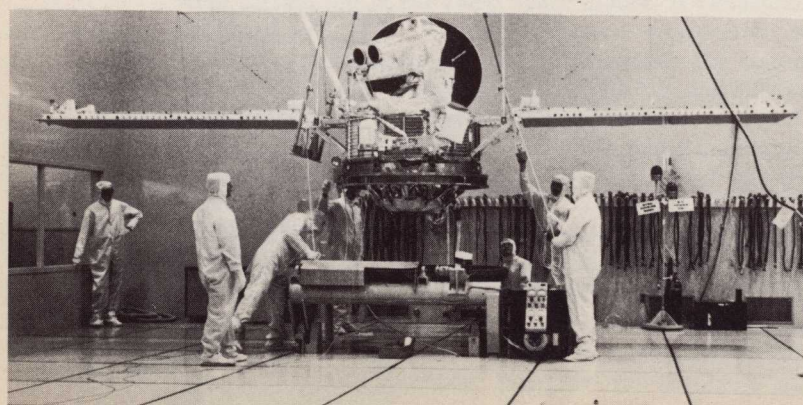
To maintain this attitude during flight required continuous control by venting minute jets of nitrogen through reaction control jets mounted at the ends of the solar panels and on the antennae and magnetometer outriggers. Two bearings ensured the correct attitude; one on the Sun, controlled by the Sun sensors, which kept the Z axis pointing at the Sun, the other on the bright star Canopus, monitored by the Canopus tracker.

Maintaining the vital transmission link to Earth

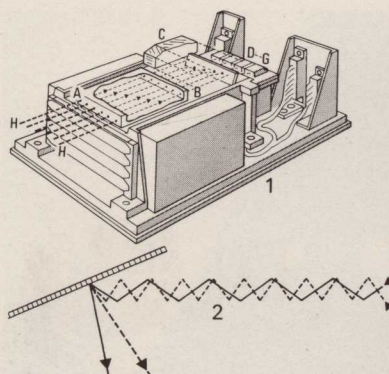
The mainstay of the temperature control was the sunshade, supplemented by multilayer thermal blankets at the top and bottom of the vehicle. Five of the eight sides were also equipped with louvred panels, which gave active temperature control by adjusting the amount of surface able to radiate into space.

The electrical power needed to operate Mariner was generated by the two solar panels carried on outriggers, which kept them clear of the shadow cast by the sunshade. A storage battery provided power for use when the solar cells were not aligned with the Sun, as happened during manoeuvres or when the Sun was occulted.

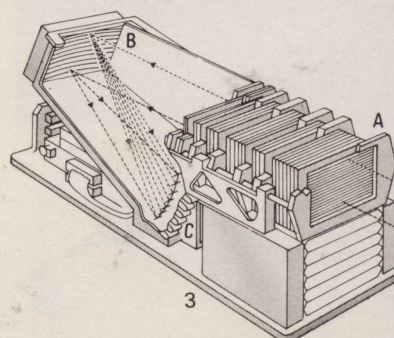
In order to function at all, Mariner had to receive commands from Earth, and transmit data back to Earth. This was achieved by the low-gain antenna and the high-gain antenna respectively; the data was received and transmitted in binary digital form.



Light in the extreme ultraviolet portion of the electromagnetic spectrum has the property of interacting strongly with most gases. Gas molecules will absorb certain ultraviolet frequencies which correspond to their characteristic wavelengths. This produces dark lines in the ultraviolet spectrum. An irradiated gas is also luminescent, emitting light at these same characteristic wavelengths. Both these effects were used to search for a Mercurian atmosphere. The occultation ultraviolet spectrometer (1) attempts to trace an atmosphere through the absorption of solar ultraviolet.



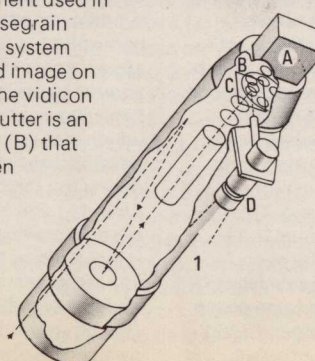
Sunlight enters the spectrometer at A and is collimated into narrow beams at B. These beams are reflected by a plane mirror (C) on which a diffraction grating is ruled. The grating splits the six beams into spectral bands (2). Detectors measure the solar ultraviolet flow at four different wavelengths centred on the absorption bands for Neon (D), Helium (E), Argon (F) and Krypton (G). Two infra-red beams (H), in the zero order reflection, time the moment of occultation.



The airglow ultraviolet spectrometer (3) detects faint light emissions from gases above the dark hemisphere. Collimation plates (A) admit sufficient light, which is reflected by a concave diffraction grating (B); this grating focuses the beam as a spectral band, which is measured by the detectors (C).



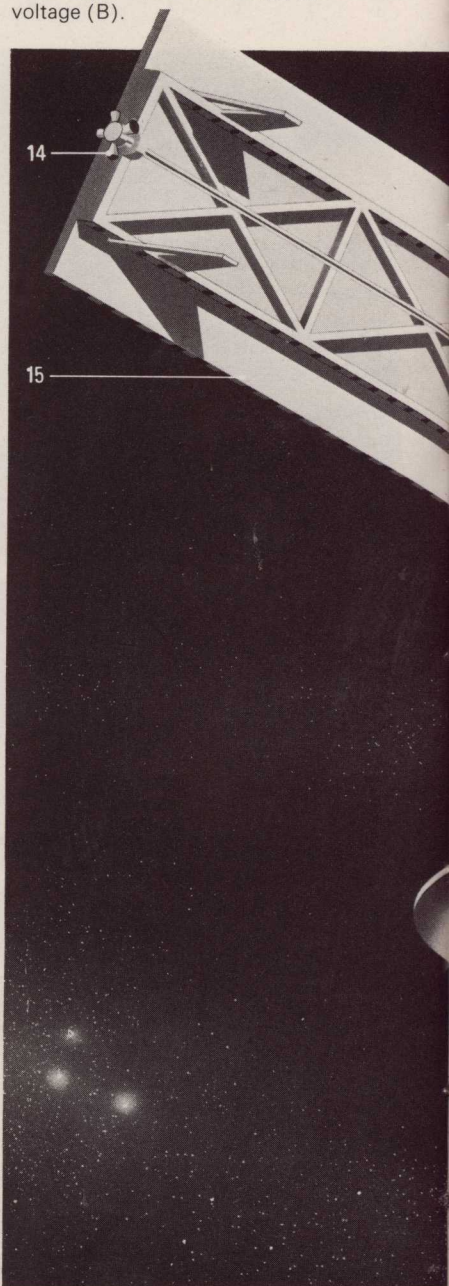
The optical equipment used in Mariner was a Cassegrain telescope (1). This optical system focuses a highly magnified image on to the sensitive screen of the vidicon tube (A). In front of the shutter is an eight-position filter wheel (B) that enables pictures to be taken at different wavelengths, including ultraviolet. The filter wheel also carries a diagonal mirror (C) that can switch the camera to a wide-angle lens system (D). As in a conventional television camera,



exposure of the vidicon screen to light produces a pattern of variable charge, proportional to the light intensity at each point. The pattern is scanned by an electron beam in a format of 700 lines, compared to 625 lines in a typical domestic picture. It is then transmitted to Earth in digital form as a series of binary numbers. A short burst at full power denotes 1; no transmission at all denotes 0.



Mariner 10's radiometer measured the temperature of the surface layers of Mercury by sensing the radiation emitted. It was accurate to within ½°C. The radiometer works by measuring the minute changes of temperature in a sensitive element placed at the focal point of a telescope (A). The sensitive element is usually a thermocouple (the junction of two dissimilar metals), which generates a voltage dependent upon its temperature. For the Mariner radiometer several junctions were joined in series to form "thermopiles", which multiply the voltage (B).



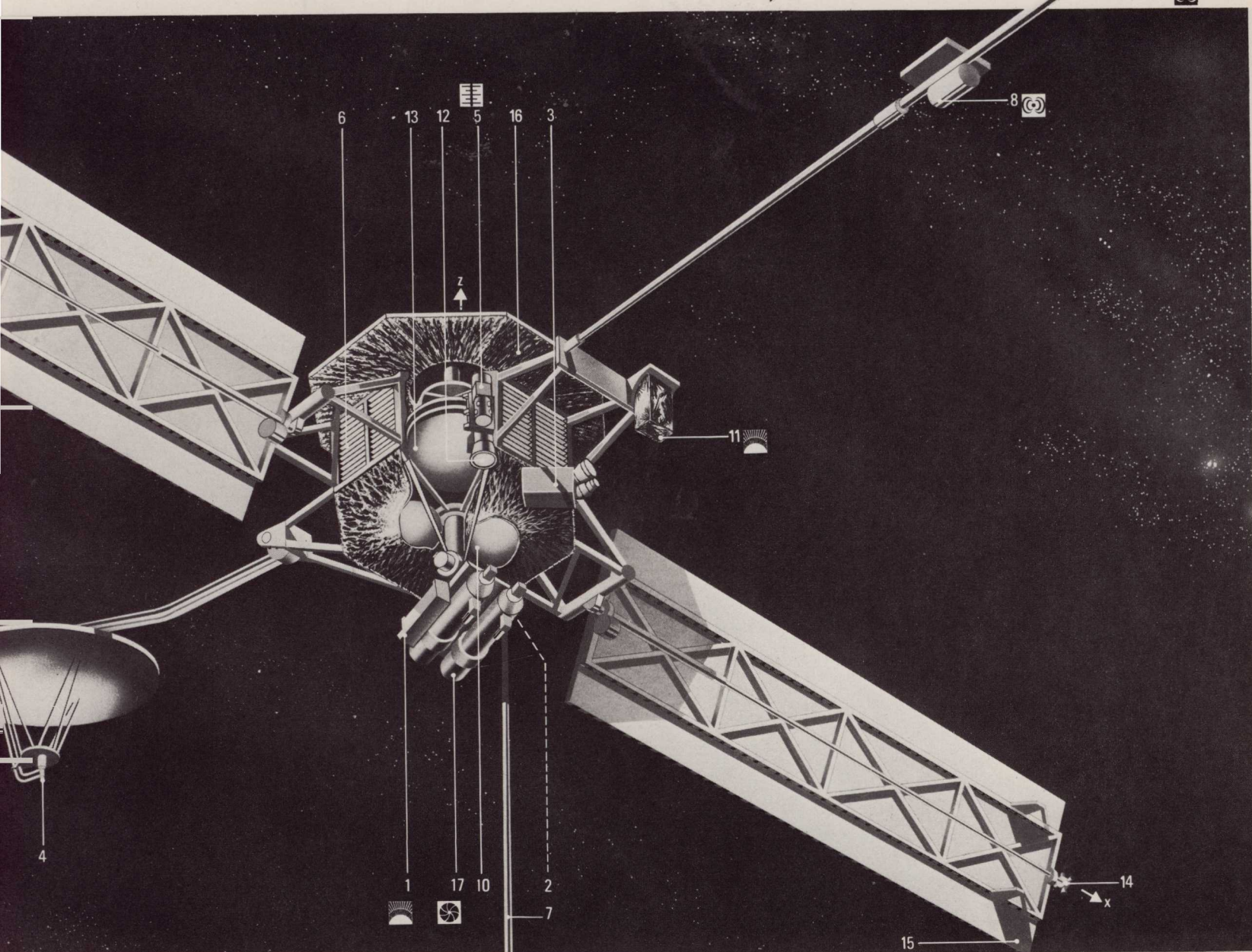
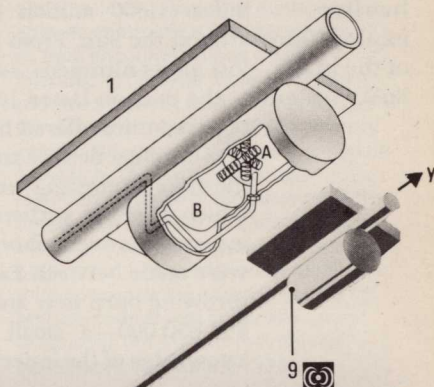


Two separate telescopes and thermopiles were used, one to measure the warm regions, the other to measure the cold. Both are mounted side by side and face a diagonal mirror (C) driven by a motor (D). Through this, mirror the telescopes view in succession an internal calibration surface at a known temperature, empty space and the surface of the planet in order to eliminate inconsistencies.



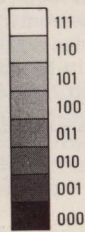
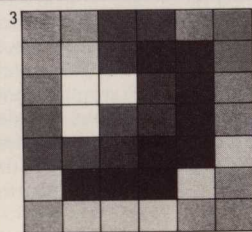
The magnetic fields of both Venus and Mercury were measured using the two magnetometers (1); because the magnetic environment continues between the planets, they were kept in operation during the whole flight. Each magnetometer consists of three measuring coils mounted at right angles to one another (A). Each coil measures the field strength along its own axis. By vector addition of the three components, the external field strength and direction can be reconstructed (2). Two magnetometers are used so that by comparing them both, the spacecraft's

own magnetic field is eliminated. To ensure the readings are constant, the coils are turned through 180° periodically. This operation is carried out with the flipper motor (B) for the X and Z coils, and by rolling the whole vehicle for the Y coils. In all 100 readings a second were made and transmitted back to Earth.



2

101	101	001	001	100	100
101	110	001	000	000	100
100	111	111	001	000	101
011	111	010	001	000	101
010	010	001	000	000	110
110	000	000	000	110	101
101	110	110	110	101	101



The electron charge on the vidicon screen is sampled 832 times for each scan line. Each sample constitutes a picture element or "pixel". The light intensity for each pixel is then converted into an eight-digit binary number.

Each pixel defines varying shades of grey from black (00000000) to white (11111111). An example (2) of the binary notation uses the first three digits of each pixel. Only six tones of grey are reproduced out of the possible 256 (3).

- 1 Airglow spectrometer
- 2 Canopus tracker
- 3 Charged particle telescope
- 4 High-gain antenna
- 5 Infra-red radiometer
- 6 Louvred panel
- 7 Low-gain antenna
- 8 Magnetometer (inboard)
- 9 Magnetometer (outboard)
- 10 Nitrogen reaction gas chamber
- 11 Occultation spectrometer
- 12 Plasma science
- 13 Propulsion unit
- 14 Reaction control jet
- 15 Solar panel
- 16 Sunshade
- 17 Television camera

- Magnetic field experiment
- Temperature measurement
- Atmospheric detection
- Photographic coverage

The Canopus tracker is located out of sight in the diagram, on the panel between the low- and high-gain antennae. The x, y and z axes are indicated at the ends of the solar panel, magnetometer boom and sunshade respectively. The z axis points towards the Sun.

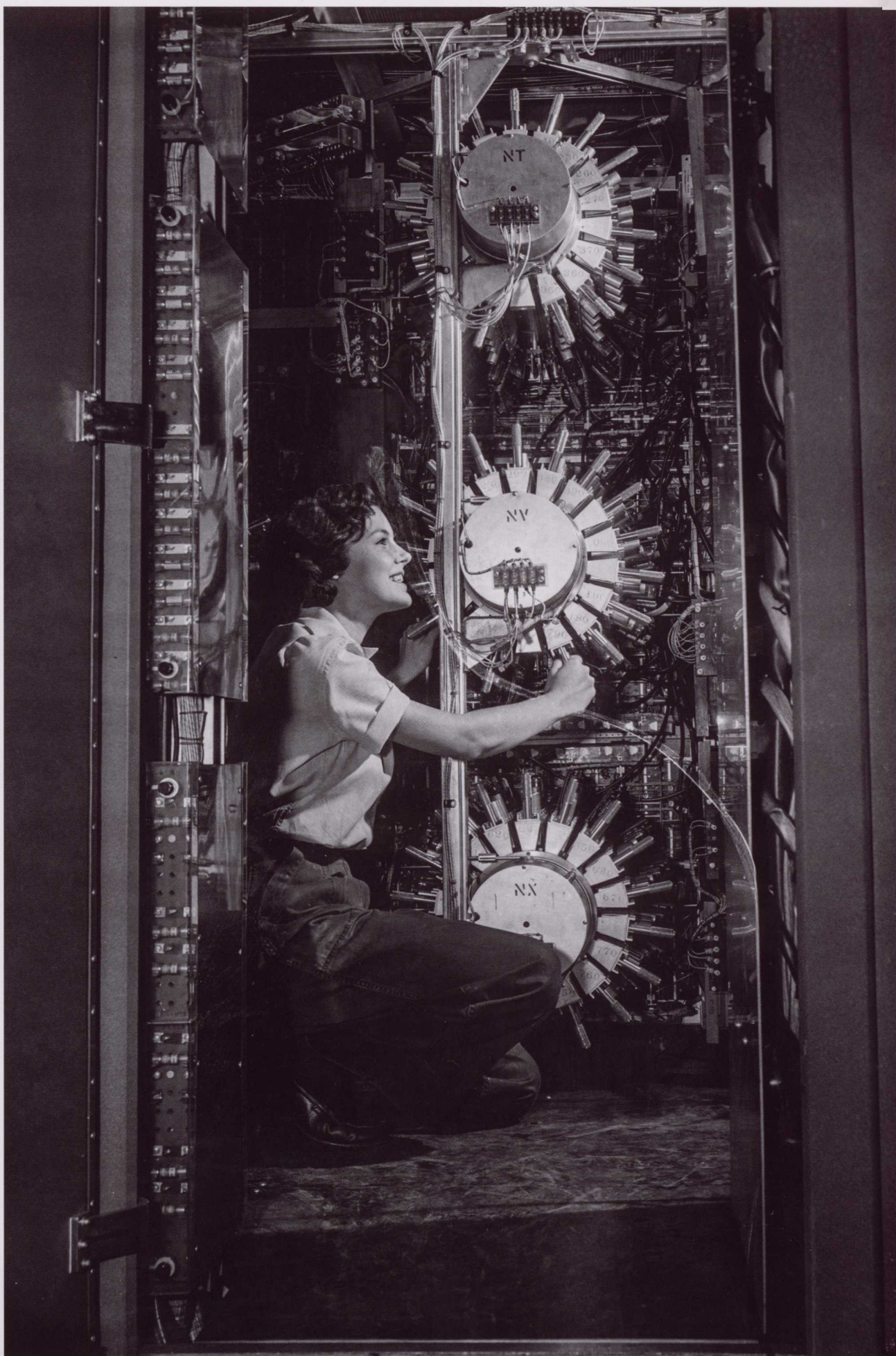
UNIVAC I was developed by John Presper Eckert (1919–1995) and John William Mauchly (1907–1980) and launched by the business-machine manufacturers Remington Rand. In May 1951, the first UNIVAC was delivered to the U.S. Census Bureau, thus becoming the United States' first commercial computer. Other early customers to order the 7.6-ton (7 tn.sh) mainframe included the U.S. Air Force, General Electric, and the Westinghouse Electric Corporation. At today's currency rates, a UNIVAC computer from the first series cost around \$7 million.

Der UNIVAC I wurde von John Presper Eckert (1919–1995) und John William Mauchly (1907–1980) entwickelt und von dem Unternehmen Remington Rand auf den Markt gebracht. Im Mai 1951 kaufte die Bundesbehörde United States Census Bureau das erste Gerät, das damit der erste kommerzielle Computer der Vereinigten Staaten war. Weitere frühe Kunden, die den 7,6 Tonnen schweren Großrechner bestellten, waren die US Air Force, General Electric und die Westinghouse Electric Corporation. Nach heutiger Umrechnung kostete ein UNIVAC-Rechner der ersten Baureihe etwa 7 Millionen US-Dollar.

L'UNIVAC I a été mis au point par John Presper Eckert (1919–1995) et John William Mauchly (1907–1980), puis commercialisé par Remington Rand. Le premier appareil est acheté en mai 1951 par le Bureau du recensement des États-Unis. C'est le premier ordinateur commercial du pays. Les autres premiers clients du supercalculateur de 7,6 tonnes comptent l'U.S. Air Force, General Electric et Westinghouse Electric Corporation. Au tarif actuel, un UNIVAC de la première série coûterait environ 7 millions de dollars.

"The first step in a study of electronic computers is to survey the areas of business operations wherein a computer may become a useful managerial tool. These areas are called data processing areas. In its day-to-day functioning, a manufacturing concern is composed of myriad channels through which money and material flow in fulfillment of the company's obligations to its stockholders, employees, vendors, customers and the government. From a data processing point of view, these areas are concerned with management's attempts to record, measure and effectively control this flow."

Basic Programming: UNIVAC File Computer Model 1, 1958



Computer technician Joyce Cade working on a UNIVAC computer at the United States Census Bureau in 1954.